Set 1: Mirror reflections: what a reflection must show

Local outcome LOGIC.SPATIAL.01: Decide whether an image is consistent with a reflection in a horizontal or vertical mirror line

Puzzle set · genre: reflection and symmetry · MP7, MP6

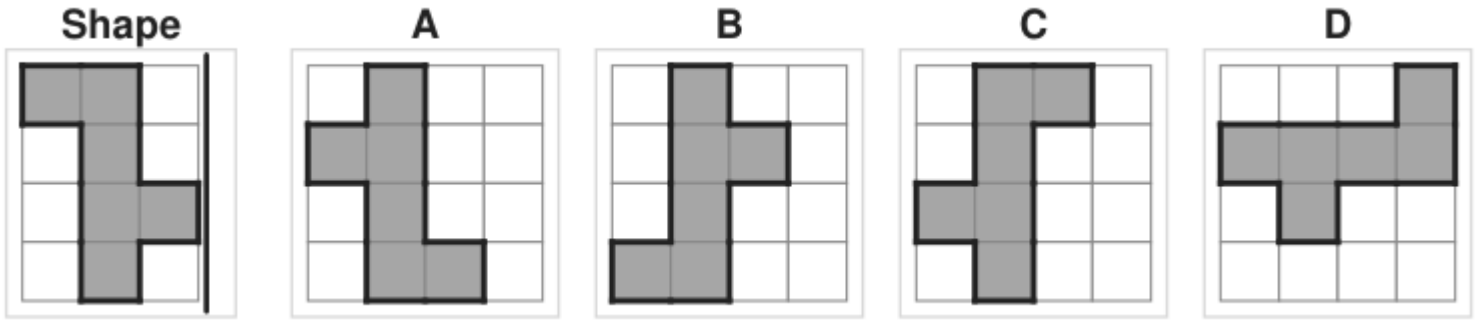
MAIN: Decide what a mirror reflection must show

BACKWARD 1: Strategy: reflect one cell at a time

FORWARD 1: ADVANCED APPLICATION: Combine reflections and read mirrored clocks and words

FORWARD 2: EXPERT CHALLENGE: Construct and count symmetric figures

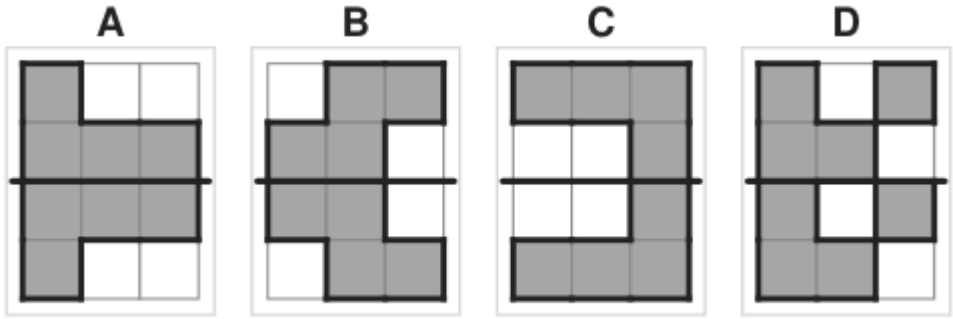
The shaded shape is reflected in the dashed mirror line on its right. Which figure shows its mirror image? (Only the shape matters, not where it sits in its box.)



A. Figure A
B. Figure B
C. Figure C
D. Figure D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q1
Logic, Set 1: Mirror reflections: what a reflection must show · MAIN · Q1/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Each figure shows a shape above a dashed water line and, below the line, what is supposed to be its reflection in the water. In which figure can the lower shape NOT be the reflection of the upper one?



A. Figure A
B. Figure B
C. Figure C
D. Figure D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q2
Logic, Set 1: Mirror reflections: what a reflection must show · MAIN · Q2/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Words are written left to right in block capitals and seen in a left-right mirror (a vertical mirror line). In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. Which word looks exactly the same in the mirror?

A. TOOT
B. ZOOM
C. NOON
D. BIB

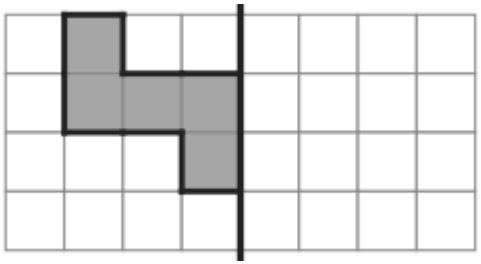
REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q3
Logic, Set 1: Mirror reflections: what a reflection must show · MAIN · Q3/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A word is written in block capitals and seen reflected in calm water below it (a horizontal mirror line). In the block capitals used here, exactly these letters look the same in a top-bottom mirror: B C D E H I K O X. The letter order stays the same in this reflection. Which word looks exactly the same in the water?

A. BEAD
B. CODE
C. CORE
D. BOXES

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q4
Logic, Set 1: Mirror reflections: what a reflection must show · MAIN · Q4/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

The grid has 4 rows and 8 columns. The dashed mirror line runs between columns 4 and 5. The shaded cells are in row 1 column 2, row 2 columns 2, 3 and 4, and row 3 column 4. Mark the cells of their mirror image. (Cells are named row,column; row 1 is the top row and column 1 the left column.)



Mark cells on the 4 × 8 grid (on screen: click each cell; cells are named row,column).

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-M-Q5
Logic, Set 1: Mirror reflections: what a reflection must show · MAIN · Q5/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A grid has 10 columns. A vertical mirror line lies between columns 5 and 6. A shaded cell is in column 2. In which column is its mirror image?

Column: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q1
Logic, Set 1: Mirror reflections: what a reflection must show · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A grid has 8 rows. A horizontal mirror line lies between rows 4 and 5. A shaded cell is in row 1. In which row is its mirror image?

Row: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q2
Logic, Set 1: Mirror reflections: what a reflection must show · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A reflection in a vertical mirror line keeps every cell in its own row.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q3
Logic, Set 1: Mirror reflections: what a reflection must show · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A cell is 3 columns to the left of a vertical mirror line. Where is its mirror image?

A. 6 columns to the right of the line
B. on the mirror line itself
C. 3 columns to the left of the line
D. 3 columns to the right of the line

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q4
Logic, Set 1: Mirror reflections: what a reflection must show · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A row of 6 cells has a mirror line between cells 3 and 4. Cell 1 is shaded. Mark the cell of its mirror image. (The grid is 1 row by 6 columns; cells are named row,column.)

Mark cells on the 1 × 6 grid (on screen: click each cell; cells are named row,column).

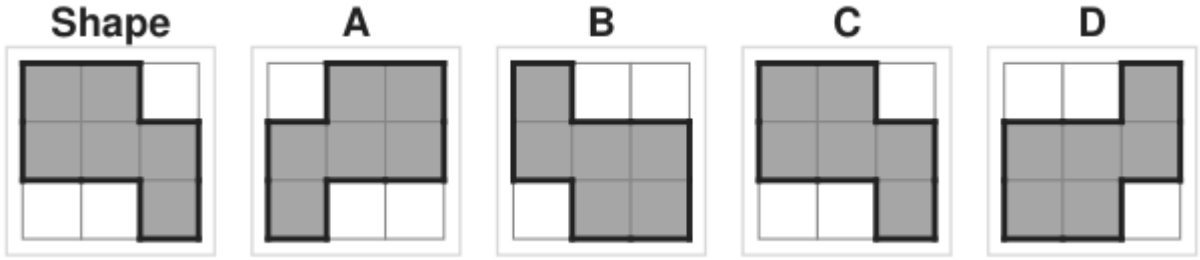
REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-B1-Q5
Logic, Set 1: Mirror reflections: what a reflection must show · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A clock face with no numbers shows 2:20. You see it in a left-right mirror. What time does the mirrored clock appear to show?

A. 10:20
B. 2:40
C. 9:40
D. 3:40

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q1
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

The shape is reflected in a vertical mirror line and then the result is reflected in a horizontal mirror line. Which figure shows the final result? (Only the shape matters, not where it sits.)



A. Figure A
B. Figure B
C. Figure C
D. Figure D

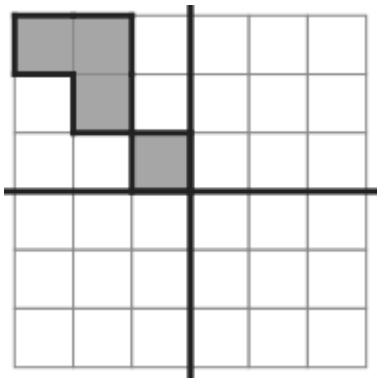
REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q2
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A word is written DOWN the page, one block capital under another, and seen in a left-right mirror. In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. Which word looks exactly the same?

A. MATH
B. MAZE
C. BATH
D. MASK

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q3
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A 6 by 6 grid has two dashed mirror lines: a vertical one between columns 3 and 4 and a horizontal one between rows 3 and 4. The figure must look the same in BOTH mirrors. Four cells are already shaded in the top-left quarter: (1,1), (1,2), (2,2) and (3,3). Mark every shaded cell of the completed figure, including the four given cells.



Mark cells on the 6 × 6 grid (on screen: click each cell; cells are named row,column).

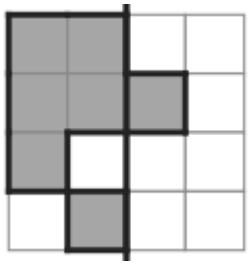
REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q4
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

A figure looks the same after a reflection in a vertical line through its center and after a reflection in a horizontal line through its center. Then it also looks the same after a half-turn about its center.

A. True
B. False

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F1-Q5
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

The figure is on a 4 by 4 grid with a dashed vertical line between columns 2 and 3. Shaded cells can be added but not removed. What is the least number of cells you must add so that the figure looks the same in a mirror along the dashed line?



Cells to add: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q1
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Strings of 3 letters are made from the letters A, H, M, O and T (letters may repeat). Each is written left to right in block capitals. In the block capitals used here, exactly these letters look the same in a left-right mirror: A H I M O T U V W X Y. How many of these strings look exactly the same in a left-right mirror?

Number of strings: _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q2
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Two vertical mirror lines are 3 cells apart: line 1 on the left, line 2 on the right. A shape is reflected in line 1 and the image is then reflected in line 2. Where does the final image end up?

A. a mirror image of the shape, 6 cells to the right
B. back exactly where the shape started
C. the original shape moved 3 cells to the right, not flipped
D. the original shape moved 6 cells to the right, not flipped

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q3
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

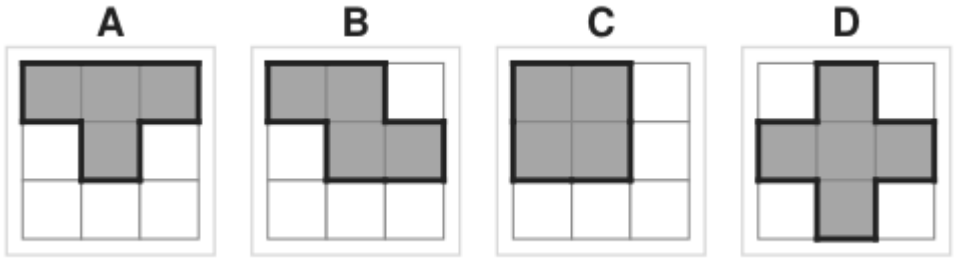
A row of 12 cells is numbered 1 to 12. A mirror line lies between cells 6 and 7. Give two cell numbers a and b, with a less than b, that are mirror images of each other.

a = _____

b = _____

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q4
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6

Count only mirror lines that run vertically, horizontally or diagonally through the center of the shape (the middle of the smallest rectangle around it). Which figure has exactly ONE mirror line?



A. Figure A
B. Figure B
C. Figure C
D. Figure D

REFERENCE LABEL: NOT PART OF THE QUESTION ID S1-F2-Q5
Logic, Set 1: Mirror reflections: what a reflection must show · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SPATIAL.01 (local) · MP7, MP6